Equity Depreciation Absorption and the Cross Section of Stock Returns

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October 20, 2025

Abstract

This paper studies the asset pricing implications of Equity Depreciation Absorption (EDA), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023a). A value-weighted long/short trading strategy based on EDA achieves an annualized gross (net) Sharpe ratio of 0.50 (0.44), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 22 (21) bps/month with a t-statistic of 2.74 (2.74), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (1 year), Net Payout Yield, Share issuance (5 year), Growth in book equity, Change in equity to assets, change in net operating assets) is 16 bps/month with a t-statistic of 2.18.

1 Introduction

Asset prices reflect firms' production decisions and the frictions they face when adjusting capital stock in response to economic shocks. The cross-section of stock returns reveals systematic patterns linked to how firms absorb depreciation costs and manage their productive capacity over time. Despite extensive research on production-based asset pricing models, we lack comprehensive evidence on how equity-based measures of depreciation absorption relate to expected returns through the lens of firms' optimal investment decisions. This gap limits our understanding of how depreciation policies and capital adjustment costs jointly determine risk premia in production economies.

In a production-based framework, firms facing higher marginal costs of capital adjustment earn higher expected returns to compensate investors for bearing systematic risk (Cochrane and Piazzesi, 2005). Equity Depreciation Absorption (EDA) captures firms' ability to internally finance replacement investment through retained earnings, directly mapping to the shadow value of installed capital in models with investment frictions (Zhang, 2005). When firms exhibit low EDA, indicating limited internal resources for depreciation coverage, they face binding financing constraints that amplify their exposure to aggregate productivity shocks through the time-varying price of capital (Liu et al., 2009).

The production function's curvature and adjustment cost specifications determine how depreciation absorption translates into conditional risk premia. Firms with negative EDA values must rely on external financing for replacement investment, exposing them to fluctuations in the stochastic discount factor through costly external finance frictions (Gomes and Liu, 2012). These firms' investment policies become more procyclical, as they delay capital replacement during downturns when the marginal product of capital is low relative to adjustment costs. This mechanism generates cross-sectional variation in expected returns that aligns with q-theory predictions about the relationship between investment frictions and risk compensation.

The signal's predictive power emerges from its connection to the firm's optimality conditions for capital accumulation. High EDA firms maintain flexibility in their investment decisions, allowing them to smooth capital adjustments across business cycle fluctuations and reducing their systematic risk exposure (Barro and Ursúa, 2008). Conversely, low EDA firms face amplified exposure to aggregate shocks through the interaction of depreciation requirements and financing constraints, generating higher required returns. This heterogeneity in firms' production technologies and adjustment costs creates persistent cross-sectional differences in how depreciation absorption maps to expected returns through the stochastic discount factor.

We find strong evidence that Equity Depreciation Absorption predicts cross-sectional stock returns. A value-weighted long/short trading strategy based on EDA achieves an annualized gross Sharpe ratio of 0.50 and generates a monthly average abnormal gross return of 22 basis points relative to the Fama-French six-factor model, with a t-statistic of 2.74. The strategy's performance remains robust across different portfolio construction methodologies, with all twenty-five gross alpha specifications yielding t-statistics exceeding 2.0, and sixteen exceeding 3.0.

The economic magnitude of the EDA premium persists after accounting for transaction costs. The net Sharpe ratio equals 0.44, placing the strategy in the top 3% of documented anomalies. Among the largest stocks—those above the 80th NYSE percentile—the EDA strategy delivers an average return of 19 basis points per month with a t-statistic of 2.04, demonstrating that the effect is not confined to small, illiquid securities. The strategy's net generalized alphas remain positive across all factor model specifications, indicating that EDA expands the achievable investment frontier even for investors with access to established factors.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the EDA trading strategy maintains an alpha of 16 basis points per month with a t-statistic of 2.18. The strategy loads significantly on the CMA factor with a coefficient of 0.31 (t-statistic of 5.68), consistent with the production-based interpretation that links depreciation absorption to investment frictions. These results establish EDA as a distinct source of cross-sectional return variation that cannot be explained by existing factor models or related accounting-based signals.

This paper advances the production-based asset pricing literature by documenting a novel link between equity depreciation absorption and expected returns. While (Fama and French, 2015) incorporate investment and profitability factors motivated by dividend discount models, and (Hou et al., 2015) develop q-factors from first-order conditions of firms' investment problems, our EDA measure directly captures the interaction between depreciation requirements and internal financing capacity. Unlike the net payout yield of (Boudoukh et al., 2007) or share issuance measures in (Daniel and Titman, 2006), EDA specifically isolates the component of equity retention tied to productive capital maintenance, providing a cleaner mapping to marginal costs in production economies.

Our methodological innovation lies in constructing a signal that bridges accounting measures of capital depreciation with market-based equity valuations, following the rigorous testing protocol of (Novy-Marx and Velikov, 2023b). This approach reveals that traditional measures of external financing activities, such as those examined in (Cooper et al., 2008), fail to fully capture the risk implications of how firms manage depreciation through internal resources. The EDA signal's orthogonality to existing anomalies—maintaining statistical significance when controlling for share issuance, book equity growth, and net operating asset changes—demonstrates that depreciation absorption represents a distinct dimension of production-based risk.

The findings contribute to broader debates about the sources of cross-sectional return predictability by establishing depreciation absorption as a fundamental characteristic linked to firms' production technologies and adjustment costs. Our results suggest that investors require compensation for bearing the systematic risk associated with firms' varying abilities to internally finance capital replacement, consistent with production-based models featuring time-varying risk premia. This evidence supports interpreting certain accounting-based anomalies through the lens of rational, production-based pricing rather than behavioral biases or market frictions alone, advancing our understanding of how real investment frictions translate into equilibrium asset prices.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of equity issuance and depreciation patterns across firms.construction of our Equity Depreciation Absorption signal relies on two key accounting variables. Common stock (CSTK) represents the par or stated value of common shares outstanding as reported on the balance sheet, reflecting the book value of equity capital contributed by shareholders at issuance. Depreciation and amortization (DP) captures the annual non-cash expense that allocates the cost of tangible and intangible assets over their useful lives, representing the systematic recognition of asset consumption in the firm's operations. Both variables are measured in millions of dollars and are directly obtained from firms' reported financial statements.construct the Equity Depreciation Absorption signal by computing the year-over-year change in common stock scaled by lagged depreciation and amortization: (CSTK(t) - CSTK(t-1)) / DP(t-1). This ratio captures the relationship between new equity issuance activity and the firm's asset depreciation rate, providing insight into how firms absorb depreciation expenses through equity financing. The numerator measures the annual change in common stock, indicating net equity issuance or repurchase activity, while the denominator normalizes this change by the prior year's depreciation expense to facilitate cross-sectional comparisons. We calculate the signal using fiscal year-end values and require non-missing values for both CSTK and DP in consecutive years. To ensure meaningful ratios, we exclude observations where lagged depreciation equals zero and winsorize the signal at the 1st and 99th percentiles to mitigate the influence of outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EDA signal. Panel A plots the time-series of the mean, median, and interquartile range for EDA. On average, the cross-sectional mean (median) EDA is -0.34 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input EDA data. The signal's interquartile range spans -0.19 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the EDA signal for the CRSP universe. On average, the EDA signal is available for 6.15% of CRSP names, which on average make up 7.36% of total market capitalization.

4 Does EDA predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EDA using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EDA portfolio and sells the low EDA portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model

(FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short EDA strategy earns an average return of 0.31% per month with a t-statistic of 3.93. The annualized Sharpe ratio of the strategy is 0.50. The alphas range from 0.22% to 0.34% per month and have t-statistics exceeding 2.74 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.31, with a t-statistic of 5.68 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 541 stocks and an average market capitalization of at least $1,475 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 24 bps/month with a t-statistics of 3.13. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for sixteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 21-33bps/month. The lowest return, (21 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.65. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EDA trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-five cases.

Table 3 provides direct tests for the role size plays in the EDA strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EDA, as well as average returns and alphas for long/short trading EDA strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80$^{\text{th}}$ NYSE percentile), the EDA strategy achieves an average return of 19 bps/month with a t-statistic of 2.04. Among these large cap stocks, the alphas for the EDA strategy relative to the five most common factor models range from 15 to 20 bps/month with t-statistics between 1.55 and 2.17.

5 How does EDA perform relative to the zoo?

Figure 2 puts the performance of EDA in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the EDA strategy falls in the distribution. The EDA strategy's gross (net) Sharpe ratio of 0.50 (0.44) is greater than 90% (97%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a $1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a $1 invested in the EDA strategy (red line).[2] Ignoring trading costs, a $1 invested in the EDA strategy would have yielded $7.48 which ranks the EDA strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a $1 invested in the EDA strategy would have yielded $5.36 which ranks the EDA strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EDA relative to those. Panel A shows that the EDA strategy gross alphas fall between the 65 and 70 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43%

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of $1 in T-bills and $1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

(52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EDA strategy has a positive net generalized alpha for five out of the five factor models. In these cases EDA ranks between the 82 and 88 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does EDA add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of EDA with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EDA or at least to weaken the power EDA has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EDA conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EDA} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EDA} EDA_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EDA,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-

[3] When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EDA. Stocks are finally grouped into five EDA portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EDA trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EDA and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EDA signal in these Fama-MacBeth regressions exceed 2.34, with the minimum t-statistic occurring when controlling for Net Payout Yield. Controlling for all six closely related anomalies, the t-statistic on EDA is 1.64.

Similarly, Table 5 reports results from spanning tests that regress returns to the EDA strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EDA strategy earns alphas that range from 16-24bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.03, which is achieved when controlling for Net Payout Yield. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EDA trading strategy achieves an alpha of 16bps/month with a t-statistic of 2.18.

7 Does EDA add relative to the whole zoo?

Finally, we can ask how much adding EDA to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the EDA signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 154-anomaly combination strategy grows to $2560.30, while $1 investment in the combination strategy that includes EDA grows to $3189.92.

8 Conclusion

This study provides compelling evidence that Equity Depreciation Absorption (EDA) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that EDA-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established risk factors and related anomalies.

The key findings reveal that a value-weighted long/short strategy based on EDA delivers an annualized net Sharpe ratio of 0.44, which remains economically meaning-

[4]We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EDA is available.

ful after accounting for transaction costs. More importantly, the strategy generates statistically significant abnormal returns of 21 basis points per month (t-statistic = 2.74) relative to the Fama-French five-factor model augmented with momentum. The robustness of these results is further confirmed by the persistence of a 16 basis points monthly alpha (t-statistic = 2.18) even after controlling for six closely related strategies from the factor zoo, including various measures of share issuance, payout yield, and balance sheet growth metrics.

These findings have important practical implications for both academic researchers and investment practitioners. The economic significance and statistical robustness of EDA suggest that it captures a distinct dimension of expected returns not fully explained by existing factor models or related accounting-based signals. For practitioners, the net Sharpe ratio of 0.44 indicates that EDA-based strategies remain profitable after realistic implementation costs, making them potentially valuable additions to quantitative investment processes.

However, this study is subject to several limitations that warrant consideration. First, while we employ the Novy-Marx and Velikov (2023) protocol to ensure robustness, the analysis is confined to U.S. equity markets, and the international validity of EDA remains unexplored. Second, the economic mechanism underlying EDA's predictive power requires further theoretical development to fully understand why this signal generates abnormal returns. Third, the stability of EDA's performance across different market regimes and its behavior during extreme market conditions merit additional investigation.

Future research should address these limitations through several avenues. International studies could examine whether EDA's predictive power extends to global equity markets, particularly in regions with different accounting standards and market structures. Researchers should also investigate the economic channels through which EDA affects returns, potentially linking it to firm investment efficiency, capital allocation decisions, or information asymmetry. Additionally, examining the interaction between EDA and other corporate events, such as mergers and acquisitions or significant capital structure changes, could provide deeper insights into its information content. Finally, exploring whether EDA's effectiveness varies across different firm characteristics, industries, or market conditions would enhance our understanding of when and where this signal is most valuable for investment decisions.

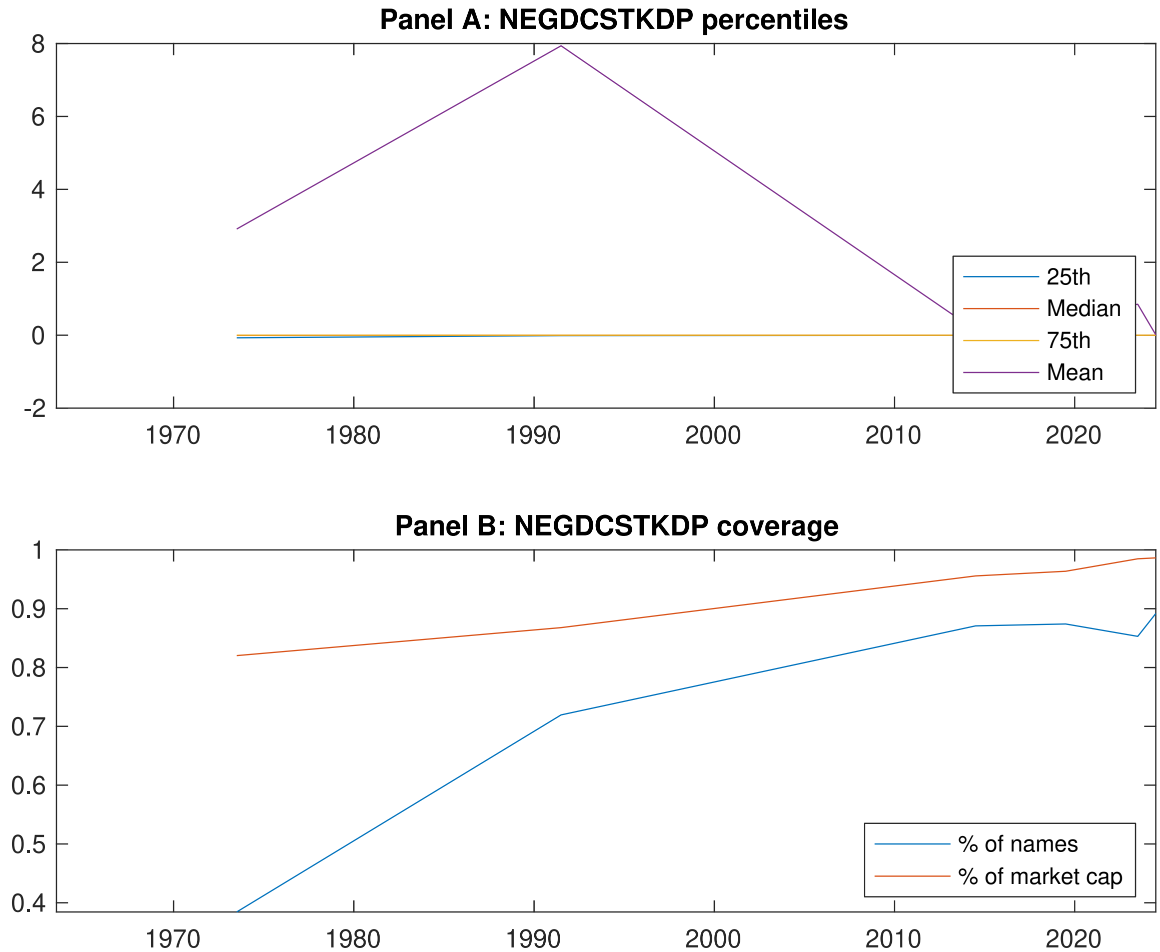


Figure 1: Times series of EDA percentiles and coverage.
This figure plots descriptive statistics for EDA. Panel A shows cross-sectional percentiles of EDA over the sample. Panel B plots the monthly coverage of EDA relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EDA. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on EDA-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.45	0.51	0.63	0.65	0.76	0.31
	[2.68]	[2.82]	[3.51]	[4.07]	[4.77]	[3.93]
α_{CAPM}	-0.12	-0.11	0.02	0.11	0.22	0.34
	[-2.37]	[-2.57]	[0.35]	[2.26]	[4.57]	[4.36]
α_{FF3}	-0.12	-0.09	0.05	0.06	0.18	0.31
	[-2.39]	[-2.12]	[0.95]	[1.43]	[4.01]	[3.96]
α_{FF4}	-0.10	-0.06	0.04	0.05	0.17	0.27
	[-1.95]	[-1.27]	[0.84]	[1.11]	[3.64]	[3.44]
α_{FF5}	-0.14	-0.05	0.06	-0.01	0.10	0.24
	[-2.71]	[-1.09]	[1.16]	[-0.16]	[2.16]	[3.05]
α_{FF6}	-0.12	-0.02	0.06	-0.01	0.09	0.22
	[-2.34]	[-0.46]	[1.05]	[-0.25]	[2.05]	[2.74]
Panel B: Fama and French (2018) 6-factor model loadings for EDA-sorted portfolios						
β_{MKT}	0.98	1.02	1.03	0.98	0.98	0.00
	[77.58]	[95.32]	[81.31]	[90.42]	[88.94]	[0.02]
β_{SMB}	-0.00	0.03	0.02	-0.02	-0.03	-0.03
	[-0.10]	[2.00]	[1.26]	[-1.54]	[-1.71]	[-0.93]
β_{HML}	0.05	-0.05	-0.09	0.03	0.02	-0.03
	[1.98]	[-2.45]	[-3.63]	[1.61]	[0.81]	[-0.85]
β_{RMW}	0.11	-0.07	-0.04	0.09	0.14	0.02
	[4.68]	[-3.39]	[-1.43]	[4.03]	[6.46]	[0.64]
β_{CMA}	-0.11	-0.08	-0.00	0.22	0.20	0.31
	[-2.97]	[-2.49]	[-0.02]	[7.01]	[6.41]	[5.68]
β_{UMD}	-0.03	-0.04	0.01	0.01	0.01	0.03
	[-2.02]	[-3.75]	[0.53]	[0.53]	[0.47]	[1.62]
Panel C: Average number of firms (n) and market capitalization (me)						
n	761	651	541	646	704	
me ($\$10^6$)	1713	1475	2038	2212	2495	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the EDA strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.31	0.34	0.31	0.27	0.24	0.22
			[3.93]	[4.36]	[3.96]	[3.44]	[3.05]	[2.74]
Quintile	NYSE	EW	0.53	0.60	0.52	0.44	0.39	0.34
			[8.38]	[9.75]	[9.13]	[7.81]	[7.24]	[6.25]
Quintile	Name	VW	0.29	0.31	0.28	0.24	0.23	0.20
			[3.65]	[3.92]	[3.46]	[2.95]	[2.83]	[2.49]
Quintile	Cap	VW	0.24	0.27	0.25	0.20	0.22	0.19
			[3.13]	[3.47]	[3.15]	[2.58]	[2.81]	[2.39]
Decile	NYSE	VW	0.37	0.39	0.33	0.27	0.30	0.25
			[4.02]	[4.22]	[3.60]	[2.86]	[3.28]	[2.71]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.27	0.31	0.27	0.25	0.22	0.21
			[3.42]	[3.97]	[3.55]	[3.27]	[2.91]	[2.74]
Quintile	NYSE	EW	0.33	0.41	0.33	0.29	0.21	0.18
			[4.73]	[6.02]	[5.25]	[4.66]	[3.51]	[3.13]
Quintile	Name	VW	0.25	0.28	0.24	0.22	0.21	0.19
			[3.16]	[3.53]	[3.07]	[2.79]	[2.64]	[2.46]
Quintile	Cap	VW	0.21	0.23	0.21	0.18	0.19	0.18
			[2.65]	[3.00]	[2.67]	[2.36]	[2.51]	[2.29]
Decile	NYSE	VW	0.33	0.35	0.29	0.26	0.27	0.25
			[3.55]	[3.77]	[3.21]	[2.82]	[2.99]	[2.70]

Table 3: Conditional sort on size and EDA

This table presents results for conditional double sorts on size and EDA. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EDA. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EDA and short stocks with low EDA .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results

		EDA Quintiles					EDA Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.48	0.69	0.91	0.96	1.00	0.55	0.60	0.53	0.47	0.40	0.36
		[1.87]	[2.68]	[3.51]	[4.04]	[4.39]	[6.35]	[6.89]	[6.46]	[5.61]	[4.94]	[4.35]
	(2)	0.54	0.72	0.88	0.98	0.90	0.37	0.43	0.34	0.30	0.29	0.26
		[2.31]	[3.01]	[3.73]	[4.22]	[4.22]	[4.16]	[4.93]	[4.05]	[3.52]	[3.37]	[3.00]
	(3)	0.61	0.61	0.76	0.86	0.93	0.33	0.37	0.30	0.26	0.25	0.22
		[2.93]	[2.87]	[3.37]	[4.23]	[4.76]	[3.98]	[4.48]	[3.78]	[3.20]	[3.10]	[2.69]
	(4)	0.49	0.62	0.83	0.73	0.81	0.32	0.35	0.28	0.28	0.12	0.14
		[2.61]	[3.08]	[4.21]	[3.80]	[4.49]	[4.01]	[4.50]	[3.76]	[3.71]	[1.71]	[1.91]
	(5)	0.49	0.50	0.48	0.58	0.68	0.19	0.20	0.18	0.15	0.18	0.15
		[2.96]	[2.83]	[2.75]	[3.51]	[4.28]	[2.04]	[2.17]	[1.95]	[1.55]	[1.89]	[1.59]

Panel B: Portfolio average number of firms and market capitalization

		EDA Quintiles Average n					EDA Quintiles Average market capitalization ($\$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	376	374	373	373	367	31	33	39	29	28
	(2)	103	101	102	102	101	55	55	56	55	54
	(3)	73	73	72	72	73	95	92	95	97	97
	(4)	62	62	62	62	62	197	201	208	206	212
	(5)	57	57	57	57	57	1436	1519	1537	1619	1891

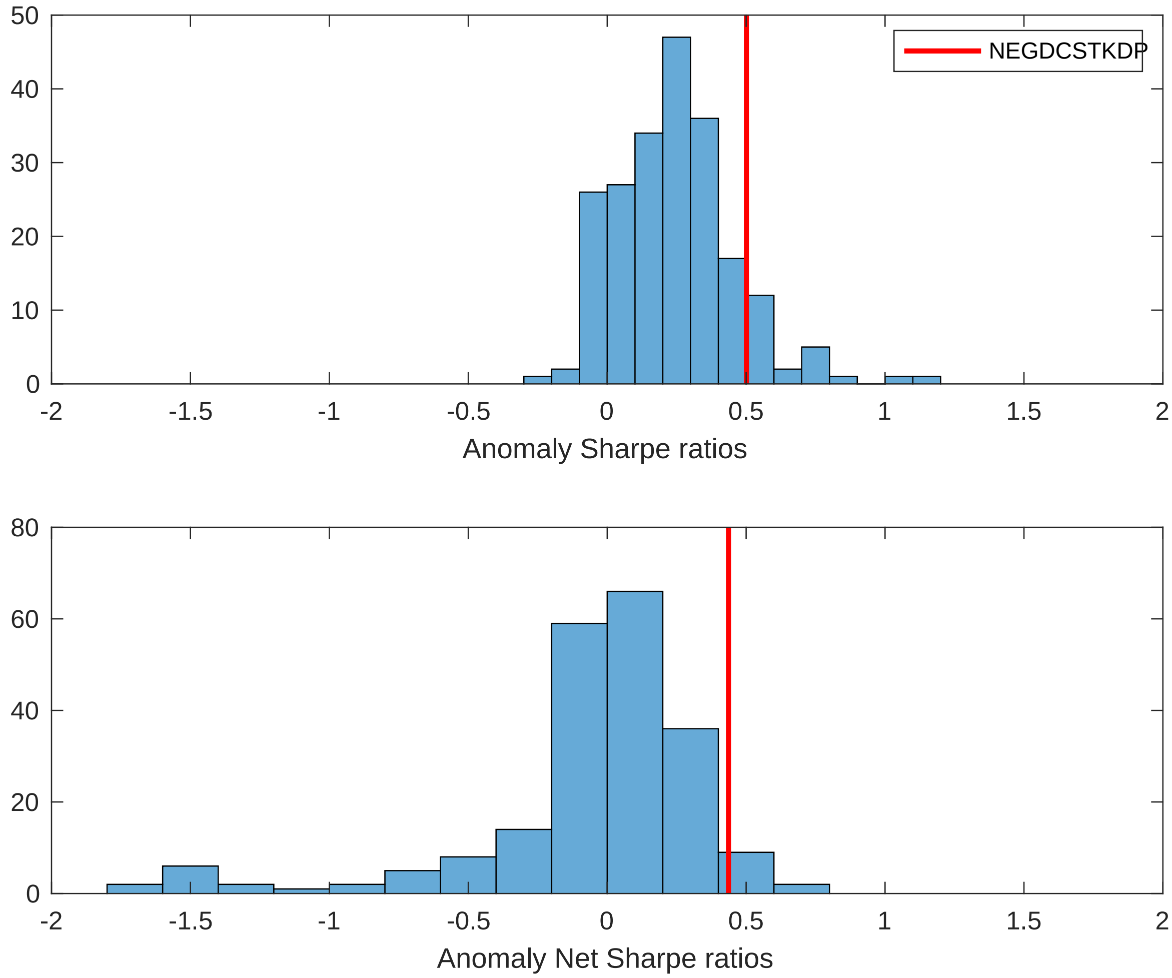


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EDA with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

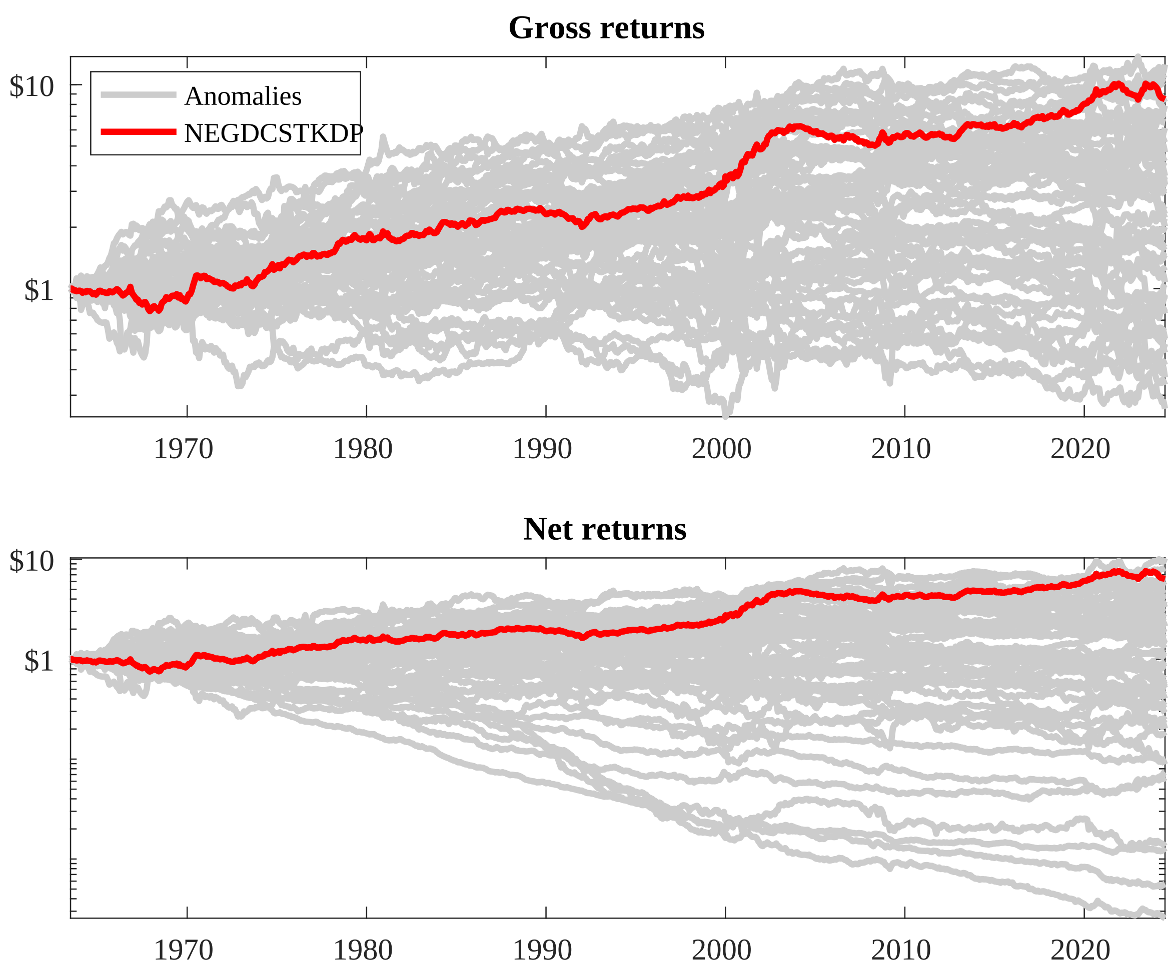
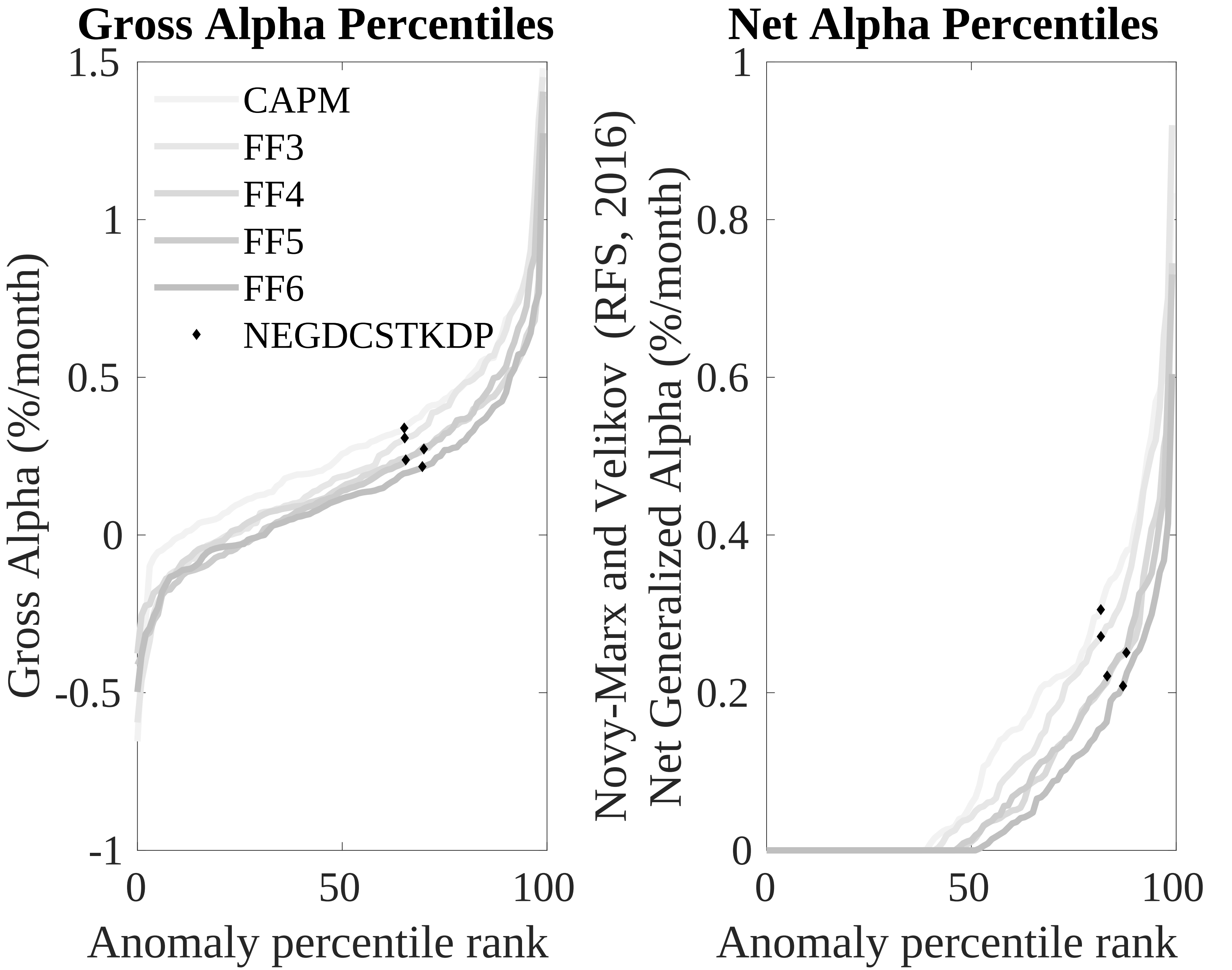
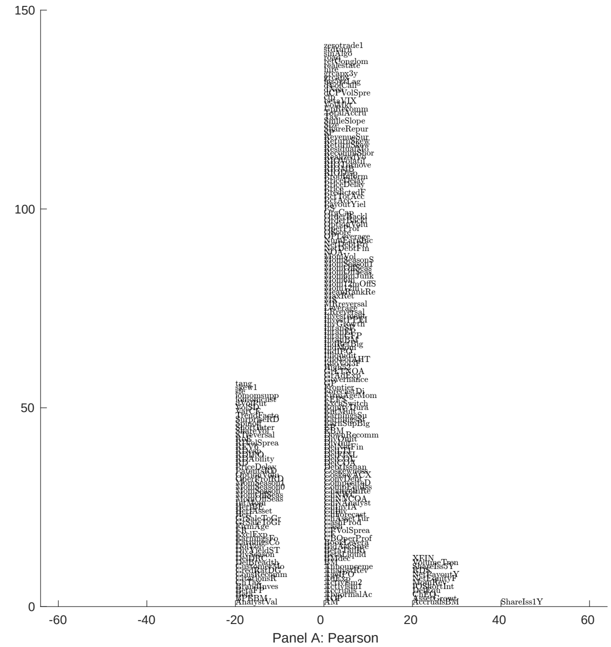


Figure 3: Dollar invested.
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EDA trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
NEGDCSTKDP
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
1
0.8
0.6
0.4
0.2
0
0
50
100
0
50
100



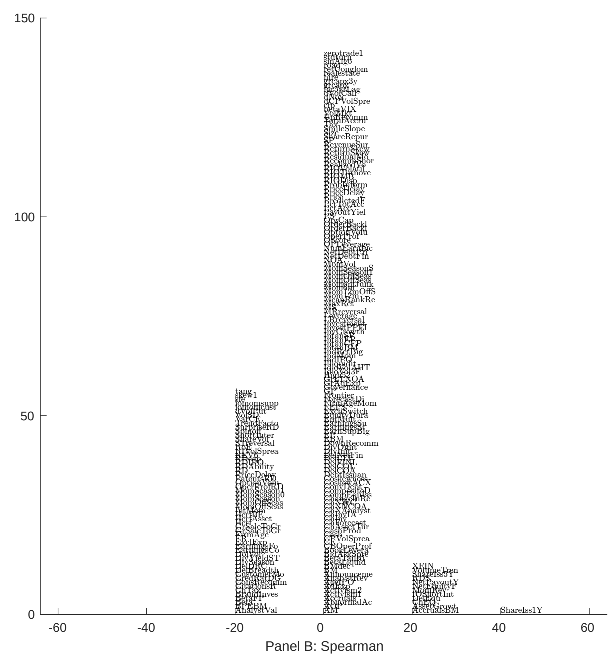


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with EDA. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

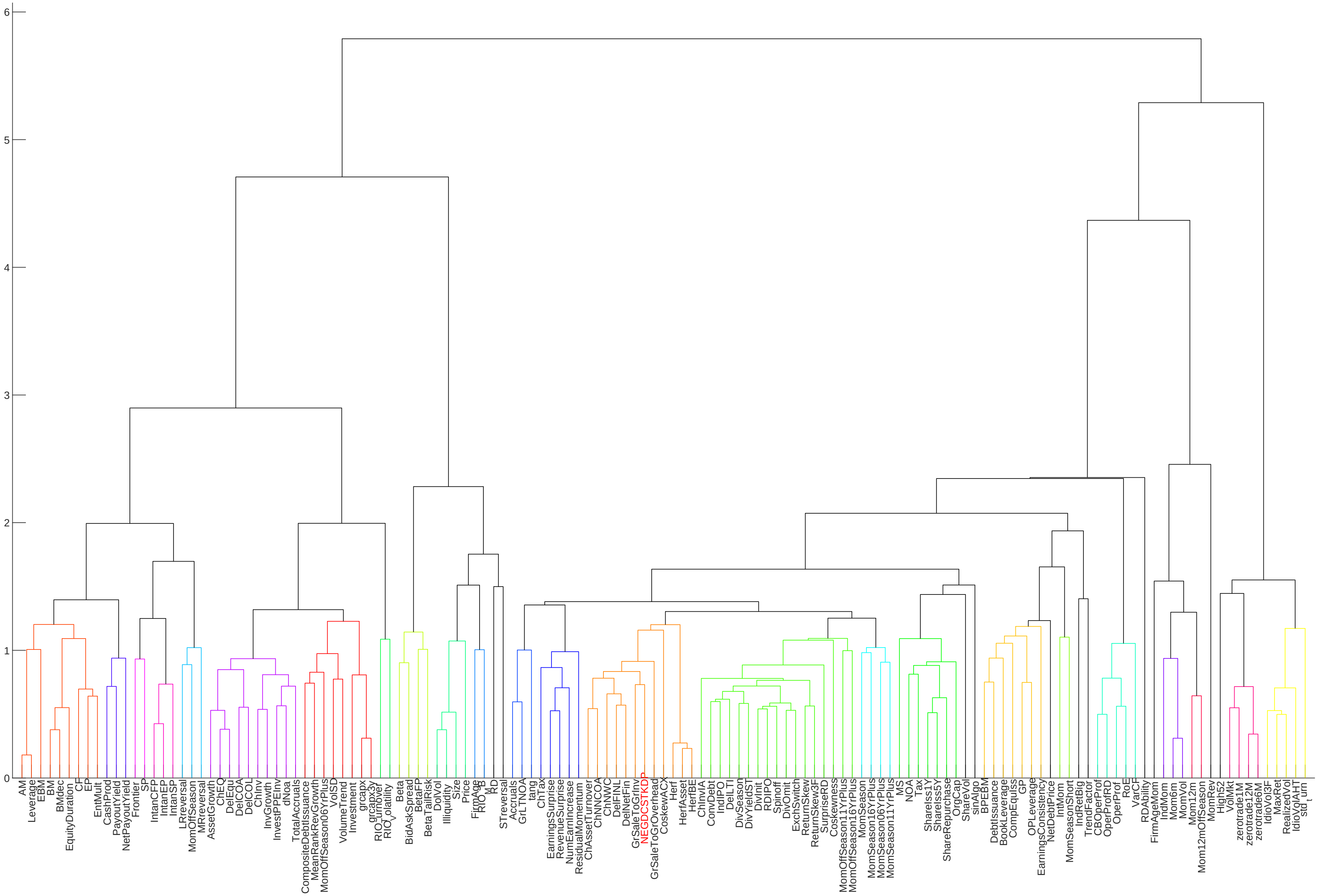


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

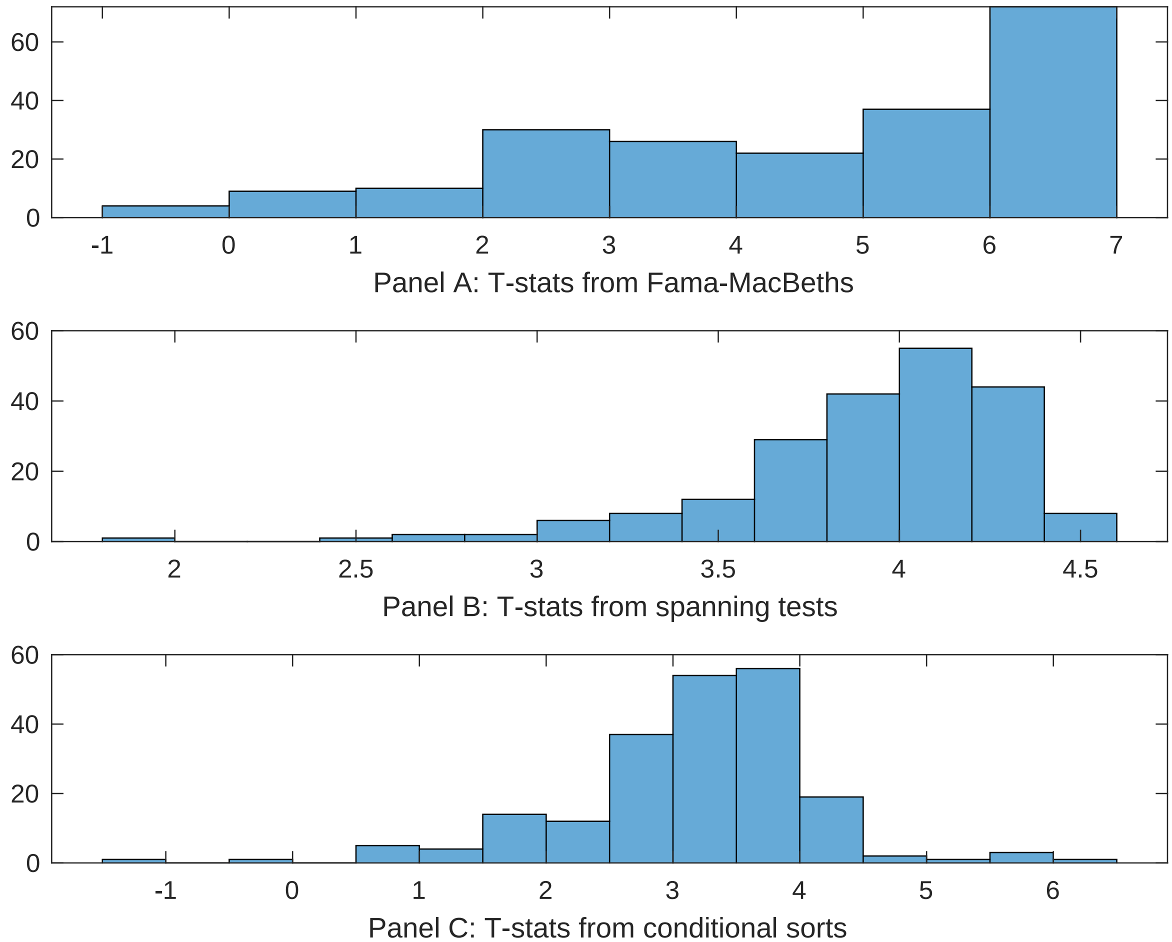


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of EDA conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EDA} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EDA} EDA_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EDA,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EDA. Stocks are finally grouped into five EDA portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EDA trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EDA. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EDA} EDA_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X, are Share issuance (1 year), Net Payout Yield, Share issuance (5 year), Growth in book equity, Change in equity to assets, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13	0.12	0.13	0.17	0.12	0.13	0.12
	[6.07]	[5.44]	[6.44]	[6.80]	[5.72]	[6.05]	[3.78]
EDA	0.12	0.74	0.11	0.12	0.13	0.12	0.51
	[5.07]	[2.34]	[4.41]	[5.36]	[5.74]	[5.09]	[1.64]
Anomaly 1	0.13						0.60
	[5.84]						[2.20]
Anomaly 2		0.21					0.10
		[1.83]					[0.98]
Anomaly 3			0.25				0.25
			[4.52]				[4.24]
Anomaly 4				0.38			-0.13
				[3.18]			[-0.56]
Anomaly 5					0.12		0.17
					[3.23]		[0.27]
Anomaly 6						0.13	0.80
						[9.00]	[5.66]
# months	715	715	715	720	720	720	715
$\bar{R}^2$(%)	0	1	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the EDA trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EDA} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Share issuance (1 year), Net Payout Yield, Share issuance (5 year), Growth in book equity, Change in equity to assets, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.21	0.24	0.16	0.21	0.22	0.21	0.16
	[2.69]	[3.13]	[2.03]	[2.73]	[2.82]	[2.68]	[2.18]
Anomaly 1	24.19						10.47
	[6.24]						[2.21]
Anomaly 2		12.77					4.31
		[4.25]					[1.30]
Anomaly 3			25.55				13.69
			[6.25]				[2.91]
Anomaly 4				25.58			31.20
				[6.19]			[5.24]
Anomaly 5					7.20		-18.70
					[1.86]		[-3.39]
Anomaly 6						6.60	2.15
						[1.50]	[0.50]
mkt	0.80	2.20	1.35	1.07	0.10	0.26	3.10
	[0.44]	[1.16]	[0.74]	[0.59]	[0.05]	[0.14]	[1.67]
smb	-0.19	0.90	0.57	-1.98	-1.22	-1.06	0.20
	[-0.07]	[0.33]	[0.21]	[-0.75]	[-0.45]	[-0.39]	[0.08]
hml	-2.11	-6.38	0.56	-4.57	-2.36	-1.92	-3.99
	[-0.60]	[-1.69]	[0.16]	[-1.29]	[-0.65]	[-0.53]	[-1.08]
rmw	-10.90	-5.22	-6.96	3.45	2.90	2.30	-11.10
	[-2.64]	[-1.30]	[-1.81]	[0.97]	[0.80]	[0.63]	[-2.53]
cma	14.40	18.63	17.04	4.58	22.11	24.44	-0.28
	[2.53]	[3.21]	[3.09]	[0.70]	[3.36]	[3.93]	[-0.04]
umd	2.19	4.31	1.62	2.85	3.49	3.18	1.20
	[1.21]	[2.34]	[0.89]	[1.58]	[1.89]	[1.72]	[0.66]
# months	728	728	728	732	732	732	728
$\bar{R}^2(\%)$	12	10	12	12	8	8	17

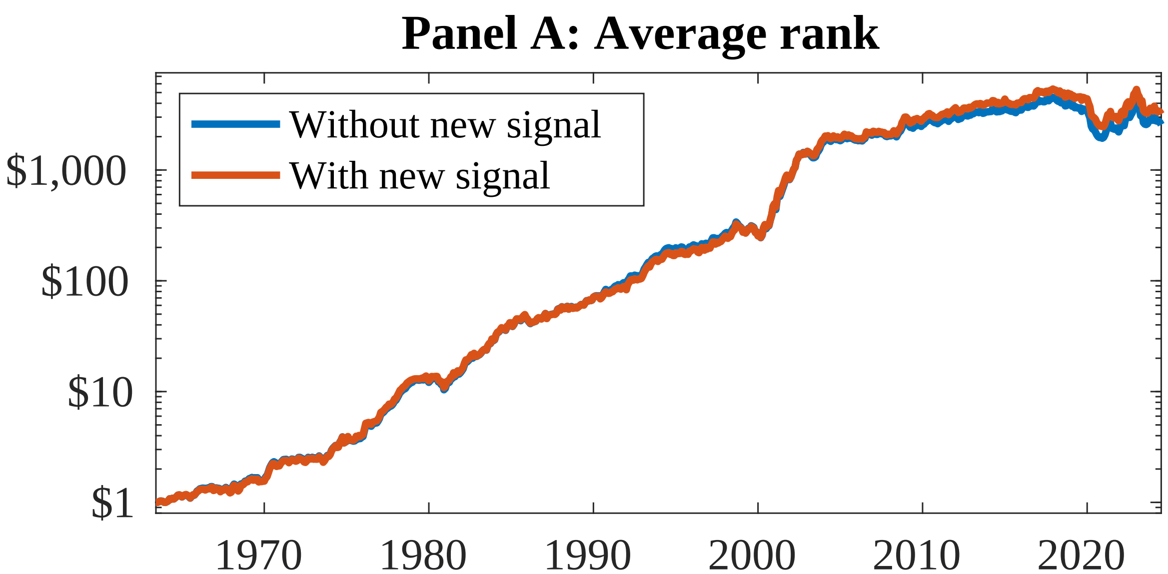


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as EDA. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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